EXPERIMENT 4

Write the Python program for the Crypt-Arithmetic problem

Register Number: **192511013**

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Aim

To write a Python program that solves the crypt-arithmetic puzzle SEND + MORE = MONEY by assigning a unique digit (0-9) to every letter so that the arithmetic addition is correct.

Algorithm

Step 1: Read the three words and collect the set of distinct letters in them. If there are more than 10 letters, the puzzle has no solution.

Step 2: Note the first letter of each word; these leading letters must not be assigned the digit 0.

Step 3: Generate every permutation of digits 0-9 taken as many at a time as there are distinct letters, and map the letters to the digits of the permutation.

Step 4: Reject the mapping if any leading letter is mapped to 0.

Step 5: Convert each word to its numeric value using the mapping (value = value x 10 + digit for each letter).

Step 6: If value(word1) + value(word2) = value(result), the mapping is a solution; stop. Otherwise try the next permutation.

Step 7: Print the letter-to-digit assignment and the solved addition. If all permutations are exhausted, print that no solution exists.

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Program

```
from itertools import permutations

def solve(word1, word2, result):
    letters = sorted(set(word1 + word2 + result))
    if len(letters) > 10:
        return None
    leading = {word1[0], word2[0], result[0]}

    def value(word, mapping):
        n = 0
        for ch in word:
            n = n * 10 + mapping[ch]
        return n

    for digits in permutations(range(10), len(letters)):
        mapping = dict(zip(letters, digits))
        if any(mapping[ch] == 0 for ch in leading):
            continue
        if value(word1, mapping) + value(word2, mapping) == value(result, mapping):
            return mapping
    return None

if __name__ == "__main__":
    w1, w2, res = "SEND", "MORE", "MONEY"
    print(f"Puzzle:   {w1} + {w2} = {res}\n")
    sol = solve(w1, w2, res)
    if sol:
        print("Letter assignment:")
        for ch in sorted(sol):
            print(f"  {ch} = {sol[ch]}")
        num = lambda w: ''.join(str(sol[c]) for c in w)
        print(f"\n  {num(w1)}\n+ {num(w2)}\n-------\n {num(res)}")
    else:
        print("No solution found")
```

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Output

```
Puzzle:   SEND + MORE = MONEY

Letter assignment:
  D = 7
  E = 5
  M = 1
  N = 6
  O = 0
  R = 8
  S = 9
  Y = 2

  9567
+ 1085
-------
 10652
```

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